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A molecular docking study to understand the interaction between anti-Cancerous compounds and 12bp DNA sequences: poly (dA-dT)₁₂ and poly (dG-dC)₁₂

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Article info

Article history:

Received 15 April 2016

Revised 05 May, 2016

Accepted 20 May 2016

Available online 31 May, 2016

Keywords:

Cancer,
Phyto-chemicals,
Docking
Minor Groove

ABSTRACT

Cancer is a leading cause of death all over the world and represents a major public health burden. Cancer is an uncontrolled growth of cell. Natural products still serve as an excellent source for the discovery and development of modern drugs for cancer treatment. We found that Anthocyanins produced greater affinity for the poly (dA-dT)₁₂ and poly (dG-dC)₁₂ DNA. It binds with the DNA minor groove with interaction energy (score value) of 9604 for AT and 8724 for GC DNA and predicted to as best anti-cancerous compounds.

INTRODUCTION

Cancer is the second leading cause of death worldwide. Chemical anti-cancer drugs cause serious side effects and, at best, merely extend the patient's lifespan by a few years (1). Cancer control may therefore benefit from the potential that resides in alternative therapies. The demand to use alternative approaches to treatment of cancer is therefore escalating. There is compelling evidence from epidemiological and experimental studies that highlights importance of compounds derived from plants "photochemical" to reduce the risk of cancer (2-4). The advantage of using such compounds is their relatively non-toxic nature and availability in an ingestive form (5-10). An ideal photochemical is one that possesses anti fungal, anti bacterial as well as anti-tumor properties with minimal toxicity and has a defined mechanism of action (11). As compounds that target specific binding

pathways are identified, how a noble drug hit a specific target, can use as a good anti-cancer drug.

Here, in this paper, we are proposing the application of molecular docking technique to find potential plant anti cancerous compounds. Reports have been available about using several molecular modeling tools such as docking, molecular dynamics and Gaussian describe the interaction of compounds with macromolecules or determine the properties of compounds (12-19). We explored molecular docking tool to determine the interaction of anti-cancerous plant compounds with the 12bp DNA sequences: poly (dA-dT)₁₂ and poly (dG-dC)₁₂.

MATERIALS AND METHODS

DNA sequences

The 12bp DNA sequences: poly (dA-dT)₁₂ and poly (dG-dC)₁₂ were designed by discovery studio software. The whole structure of the DNA was targeted for our molecular docking study.

Compounds library

Information about different anti-cancerous plant compounds were collected from the literature. Thanks to the Ayurvedic science and reports available for plant derived anti-tumor compounds. Total 21 anti-cancerous plant compounds were listed from the literature and their SMILES strings were obtained from PUBCHEM database. CORINA demo server (http://www.molecular-networks.com/online_demos/corina_demo) was employed in order to convert the SMILES strings of the compounds into 3D structures. Different anti-carcinoma drugs information was also obtained from the Drug bank database (www.drugbank.ca) and their 3D structures were downloaded.

Automatic Docking

The computational molecular docking method was employed for screening of anti-cancerous plant compounds against the structure of DNA. The molecule docking was performed by Patch dock software. We selected the Drug-complex platform of Patchdock for doing our computational docking. The anti-tumor compounds were sorted at the end of the docking process based on their Patchdock docking score produced by the docking via Patch dock software. The most stable conformation of the plant origin anti-cancerous compound was selected based highest score value. Furthermore, the compound binding was analyzed at the minor grove by discovery studio software.

Z score analysis

Statistical test such as Z score was performed for the interaction energies produced by Patch dock. The Z score determines the uniqueness and confidence in the docking result. The measurement of the Z score demonstrates the affinity between anti-cancer compounds and AT, GC DNA. It shows that the affinity produced by the best compound (with lowest interaction energy) for FAT10 protein is unlikely to produce by other compounds in a random population. Later the best compound was selected with based on the more negative Z score of interaction energy and P value less than < 0.05 (95% confidence interval). Z score: $Z = (x - \mu) / \sigma$.

RESULTS

Total 21 plant anti-cancerous compounds and five chemical anti-cancerous compounds were screened against the DNA structures of poly (dA-dT)₁₂ and poly (dG-dC)₁₂ via Patchdock. We found that plant anti-cancer compound Anthocyanins produced greater affinity for the DNA structures of poly (dA-dT)₁₂ and poly (dG-dC)₁₂ as compared to chemical anti-cancerous compounds. The Anthocyanins binds with both the DNA at their minor grooves with the highest patchdock docking score of 9604 for the case of poly (dA-dT)₁₂ and 8724 for poly (dG-dC)₁₂ (Table 1, Fig 1). On the other hand, the affinity produced by the chemical anti-cancerous compound Cyclophosphamide produced lower patchdock docking score of 2956 for the case of poly (dA-dT)₁₂ and 2882 for poly (dG-dC)₁₂ than plant anti-cancer compounds (Table 1, Fig 1).

Table 1: Patchdock docking score of Plant and Chemical anti-cancerous compounds with the poly (dA-dT)₁₂ and poly (dG-dC)₁₂ DNA.

Chemical compound	Patchdock score	
	poly (dA-dT) ₁₂	poly (dG-dC) ₁₂
Cyclophosphamide	2956	2882
Busulfan	2946	2538
Methotrexate	2760	2722
Iomustine	2760	2722
Carmustine	2376	2536
Cisplatin	1794	1786

Plant compound	Patchdock score	
	poly (dA-dT) ₁₂	poly (dG-dC) ₁₂
Anthocyanins	9604	8724
Azadirachtin	6008	5994
Aegicerin	4718	4696
Artonin E	4664	4664
Abbyssinone	4542	4482
Artonol B	4408	2480
Arctigenin	4026	3888
Avenanthramide	3268	3158
Angolensin	3224	3158
Aporphine	3136	3060
Acacetin	3044	2958
Arbutin	2934	2882
Apigenin	2926	2852
Alphahumulene	2886	2860
Altholactone	2780	2812
Angelicin	2398	2342
Anethole	2258	2298
Allicin	2238	2084
4chromone	2004	1998
AscorbicAcid	1998	1922
Amantidine	1780	1744

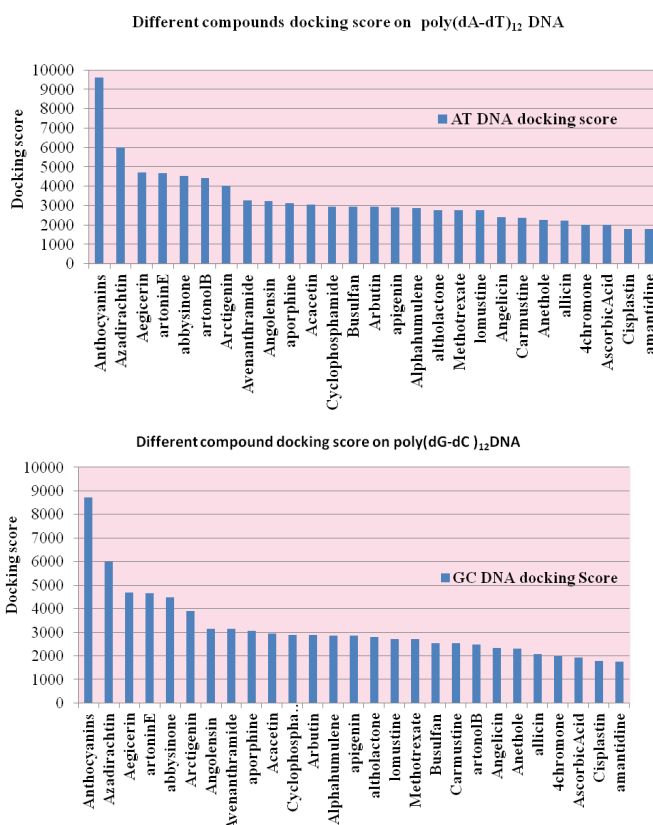


Fig 1: Distribution of Patchdock docking scores of Plant anti-cancerous compounds with the poly (dA-dT)₁₂ and poly (dG-dC)₁₂ DNA.

The Z score analysis reveals that Anthocyanin produces Z score of 3.88 with the poly (dA-dT)₁₂ and 3.72 with the poly (dG-dC)₁₂ DNA, which determines the uniqueness and confidence in the affinity of Anthocyanins for the minor grooves of both the DNA structures (Fig 2). It shows that the affinities produced by the Anthocyanins (with higher patchdock docking scores and Z scores) for the poly (dA-dT)₁₂ and poly (dG-dC)₁₂ DNA are unlikely to produce by other compounds in the random population. On the other hand, screening of chemical anti-Cancer drugs shows that no drug is having such good affinity for DNA. Also the

Z score analysis reveals the confidence on the docking results.

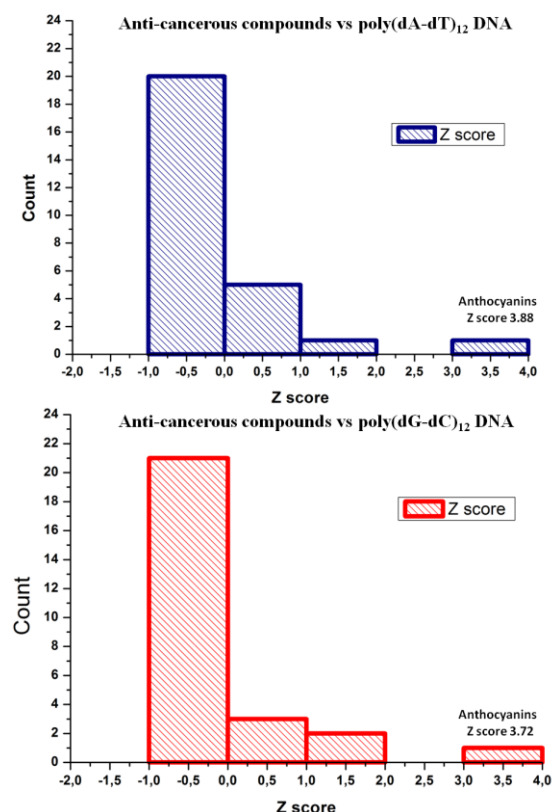


Fig 2: Anthocyanin produces unique Z score of 3.88 with the poly (dA-dT)₁₂ and 3.72 with the poly (dG-dC)₁₂ DNA

Binding site analysis shows that Anthocyanins makes hydrogen bond interactions on the minor grooves of the DNA. The long backbone of the Anthocyanins span the minor groove of the DNA and arms make hydrogen bonds with the base pairs.

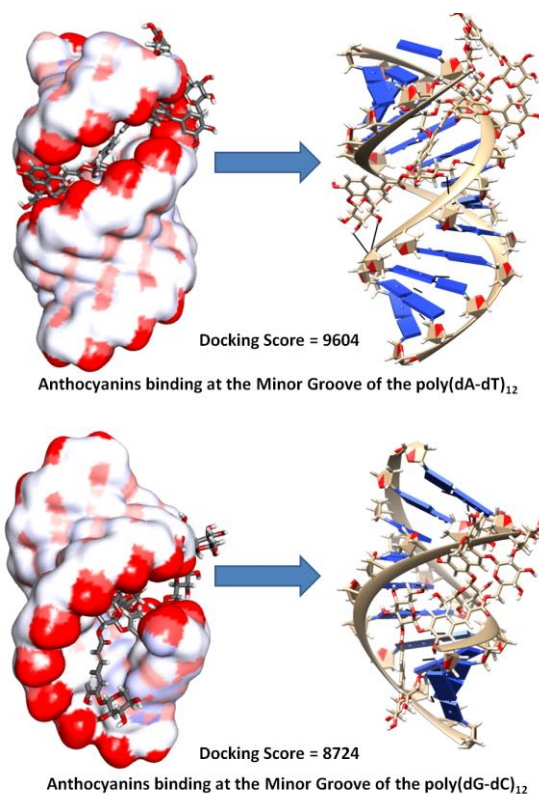


Fig 3: Anthocyanins makes hydrogen bond interactions on the minor grooves of the DNA

Anthocyanins are members of the flavonoid group of phytochemicals, a group predominant in teas, honey, wines, fruits, vegetables, nuts, olive oil, cocoa, and cereals. The colorful anthocyanins are the most recognized, visible members of the bioflavonoid phytochemicals. The free-radical scavenging and antioxidant capacities of anthocyanin pigments are the most highly publicized of the modus operandi used by these pigments to intervene with human therapeutic targets. Anthocyanin isolates and anthocyanin-rich mixtures of bioflavonoids may provide protection from DNA cleavage, estrogenic activity (altering development of hormone-dependent disease

symptoms), enzyme inhibition, boosting production of cytokines (thus regulating immune responses), anti-inflammatory activity, lipid peroxidation, decreasing capillary permeability and fragility, and membrane strengthening.

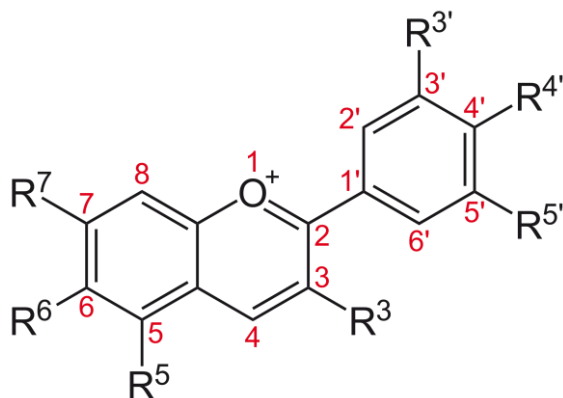


Fig 4: Anthocyanins are glycosides of anthocyanidins.

DISCUSSION

We determine the effectiveness of the computational molecular docking method to find alternative natural medicine for treating carcinoma. The patchdock is able to detect that Anthocyanin shows good affinity for DNA, better to the chemical drug Methotrexate, Busulfan, lomustine, Cyclophosphamide, Cisplatin and Carmustine. Our docking analysis also shows that Anthocyanin with higher affinity for DNA may able to inhibit the function of DNA and may help in reducing the effect of carcinoma.

CONCLUSION

Our molecular docking work concludes that Anthocyanins has compatible affinity with both the poly (dA-dT)₁₂ and poly (dG-dC)₁₂ DNA. Our computational docking work reveals that plant anti-tumor compound Anthocyanins can be useful to treat cancer. As

plant origin of Anthocyanins, therefore, it is easily available and produces no side effect. Our research work may help the medical biologist to find potential drugs from natural compounds against carcinoma.

ACKNOWLEDGEMENTS

Indu Chaturvedi greatly acknowledges Indian Science and Technology Foundation (ISTF) for providing all financial supports and computer facility for running docking calculations.

REFERENCE

1. Chaturvedi I., (2015); A Molecular Docking study to find Natural Inhibitor against FAT10 Protein for curing Hepatic Carcinoma. *J. O. Sci.* 1(2)1-9.
2. Amin A., Gali-Muhtasib H., Ocker M., Schneider-Stock R. (2009); Overview of major classes of plant derived anticancer drugs. *Int. J. Biomed. Sci.* 5 (1) 1-11.
3. Cragg G. M., and Newman D. J. (2005); Plants as a source of anti-cancer agents. *J. Ethnopharmacol.* 100: 72-9.
4. Sang U. P. (2012) Anticancer Compounds From Plants. *EXCLI Journal.* 11: 386-389.
5. Chaturvedi I., Dutta T. K., Singh P. K., and Sharma A. (2015); Effect of combined herbal feed additives on methane, total gas production and rumen fermentation. *Bioinformation.* 11 (5).
6. Chaturvedi I., Dutta T. K., and Singh P. K. (2015); Effect of different herbal feed additives on in vitro rumen fermentation. *J. O. Sci.* 1 (1).
7. Chaturvedi I., (2015); Effect of various herbal supplements on invitro rumen

- fermentation using goat rumen liquor. *J.O. Sci.* 1 (1): 24-28.
8. Chaturvedi I. and Sharma A. (2014); Inhibitor Prediction against Methenyl Tetrahydromethanopterin Enzyme from Methanobrevibacter Ruminantium via Molecular Docking to Mitigate Methane Emissions. *J. Agroecol. & Natur. Resour. Manag.* 1: 113-117.
 9. Chaturvedi I., Dutta T. K., Singh P. K., Sharma A. and Kumar M. (2014); Effect of Herbal Feed Additives on IVDMD, Methane and Total Gas Production Via in-vitro Study. *J. Agroecol. Natur. Resour. Manag.* 1: 108-112.
 10. Chaturvedi, I., Singh P. K. and Dutta T. K. (2013); Effect of Herbal Feed on Goat Haematological and Biochemical Profile. *Int. J. Biotech. Bioengi. Res.* 4: 257-262.
 11. Chaturvedi I. and Shukla K. P. (2016); Halophilic bacterial inoculants for improving plant growth and biocontrol. *J. O. Sci.* 1 (3)1-11.
 12. Sharma A., Tendulkar A. V., and Wangikar P. P. (2011); Drug Discovery against H1N1 virus (Influenza A virus) by computational virtual screening approach. *Medi. Chem. Res.* 20 (9): 1445-1449.
 13. Sharma A, and Chaudhary P. (2011); Structure modeling and inhibitor prediction of NADP oxidoreductase enzyme from Methanobrevibacter smithii. *Bioinformation*, 6 (1): 15-19.
 14. Sharma A., Tendulkar A. V., and Wangikar P. P. (2011); Structure based prediction of functional Sites with potential inhibitors to Nudix Enzymes from disease causing microbes. *Bioinformation* 5 (8): 341-349.
 15. Sharma A., and Nigam A., (2010); Structure Modeling of novel 3-methyladenine DNA glycosylase enzyme from oral pathogen Streptococcus sanguinis. *Bioinformation*, 5 (3): 136-140.
 16. Sharma A., Subbias K. K., Robine O., Chaturvedi I., Anshul N., Sharma N., and Chaudhary P. P. (2012); Computational finding of potential inhibitor for Cytochrome P450 Mono-oxygenases Enzyme of Mycobacterium tuberculosis. *Bioinformation*. 8 (19): 931-937.
 17. Scuderi D., Bakker J. M., Durand S., Maitre P., Sharma A., Martens J. K., Nicol E., Clavaguera C., Ohanessian G. (2011); Structure of singly hydrated, protonated phospho-tyrosine. *Int. J. Mass Spectrom.* 308 (2): 338-347.
 18. Semrouni D., Sharma A., Dognon J. P., Ohanessian G., Clavaguera C., (2014); Finite Temperature Infrared Spectra from Polarizable Molecular Dynamics Simulations. *J. chem. theory computat.* 10 (8): 3190-3199.
 19. Plonska-Brzezinska M. E., Bobrowska D. M., Sharma A., Rodziewicz P., Tomczyk M., Czyrko J., Brzezinski K., (2015); Triple helical collagen-like peptide interactions with selected polyphenolic compounds. *RSC Advances*. 5 (116): 95443-95453.